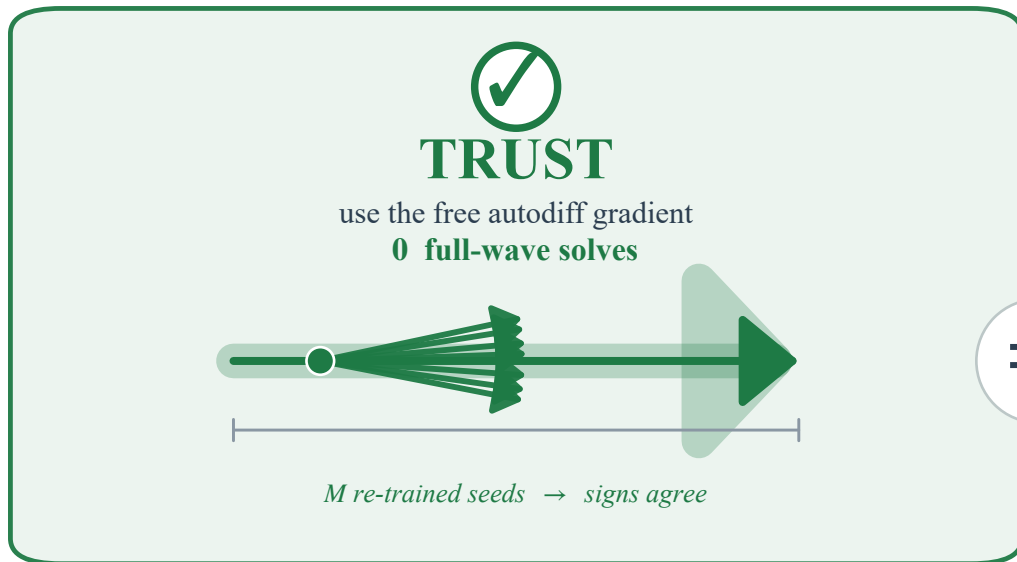
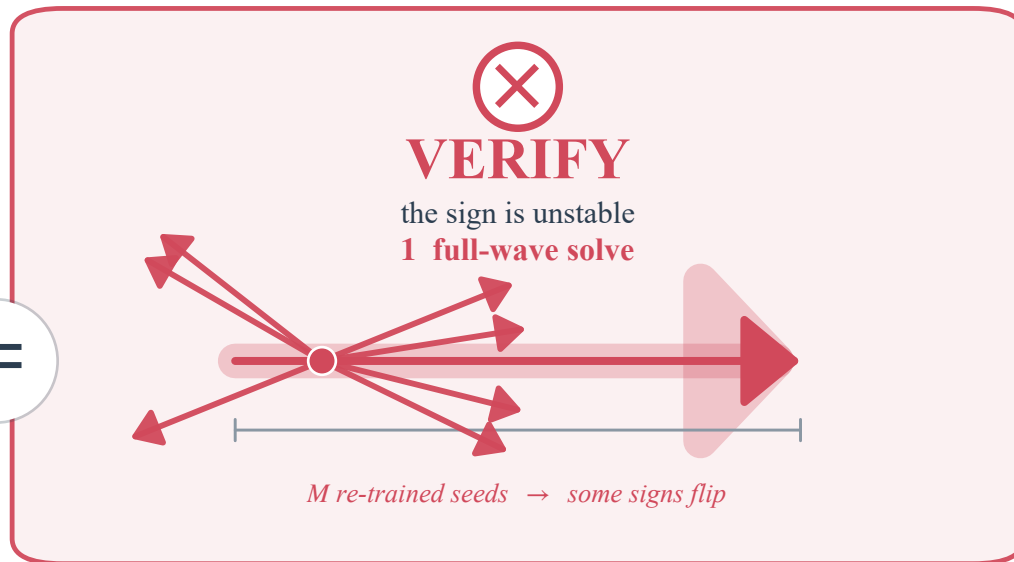


# Direction, not magnitude

*which design gradients of a differentiable surrogate to trust is set by cross-seed sign agreement, not gradient size*



=



**same size — opposite verdict**

trust the **sign agreement** across seeds, not the magnitude

**AUC 0.91**

predicts gradient  
sign-correctness



trusts the impedance gradient  
rejects the seed-unstable  
radiation gradient

**SOLVER-FREE**

the gate itself needs no full-wave solver  
*validated on an external transfer-matrix benchmark*